

Day_6__Choosing the Right Chart

September 11, 2026

0.1 Day 6 - Choosing the Right Chart

0.2 Objective

The objective of this task is to understand how to choose the appropriate chart based on the question being asked from the data.

Different chart types are useful for different purposes such as trends, comparison, distribution, relationships, frequency, correlation, and multiple relationships.

In this task, seven different chart types are used to visualize the Customer Churn Dataset.

```
[1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_excel(
    "Day_4_Ensemble_Learning_Customer_Churn.xlsx",
    sheet_name="Customer Churn Data"
)
df
```

```
[1]:
```

	CustomerID	Age	Tenure_Months	MonthlyCharges	Contract	\
0	1001	25	3	65	Month-to-month	
1	1002	34	18	72	One year	
2	1003	45	36	85	Two year	
3	1004	29	7	55	Month-to-month	
4	1005	52	48	95	Two year	
5	1006	41	24	78	One year	
6	1007	23	2	50	Month-to-month	
7	1008	38	30	88	One year	
8	1009	49	42	92	Two year	
9	1010	31	12	60	Month-to-month	
10	1011	27	8	68	Month-to-month	
11	1012	56	60	105	Two year	
12	1013	44	28	82	One year	
13	1014	36	20	75	One year	
14	1015	30	15	58	Month-to-month	
15	1016	47	40	90	Two year	
16	1017	22	4	45	Month-to-month	

17	1018	39	32	80	One year
18	1019	53	54	100	Two year
19	1020	33	16	70	Month-to-month

	InternetService	TechSupport	PaymentMethod	Churn
0	Fiber optic	No	Electronic check	Yes
1	DSL	Yes	Bank transfer	No
2	Fiber optic	Yes	Credit card	No
3	DSL	No	Electronic check	Yes
4	Fiber optic	Yes	Credit card	No
5	DSL	Yes	Bank transfer	No
6	No	No	Mailed check	Yes
7	Fiber optic	Yes	Electronic check	No
8	Fiber optic	Yes	Credit card	No
9	DSL	No	Mailed check	Yes
10	Fiber optic	No	Electronic check	Yes
11	No	Yes	Bank transfer	No
12	DSL	Yes	Credit card	No
13	Fiber optic	No	Electronic check	Yes
14	DSL	No	Mailed check	Yes
15	Fiber optic	Yes	Credit card	No
16	No	No	Mailed check	Yes
17	DSL	Yes	Bank transfer	No
18	Fiber optic	Yes	Credit card	No
19	DSL	No	Electronic check	Yes

0.3 1. Line Chart

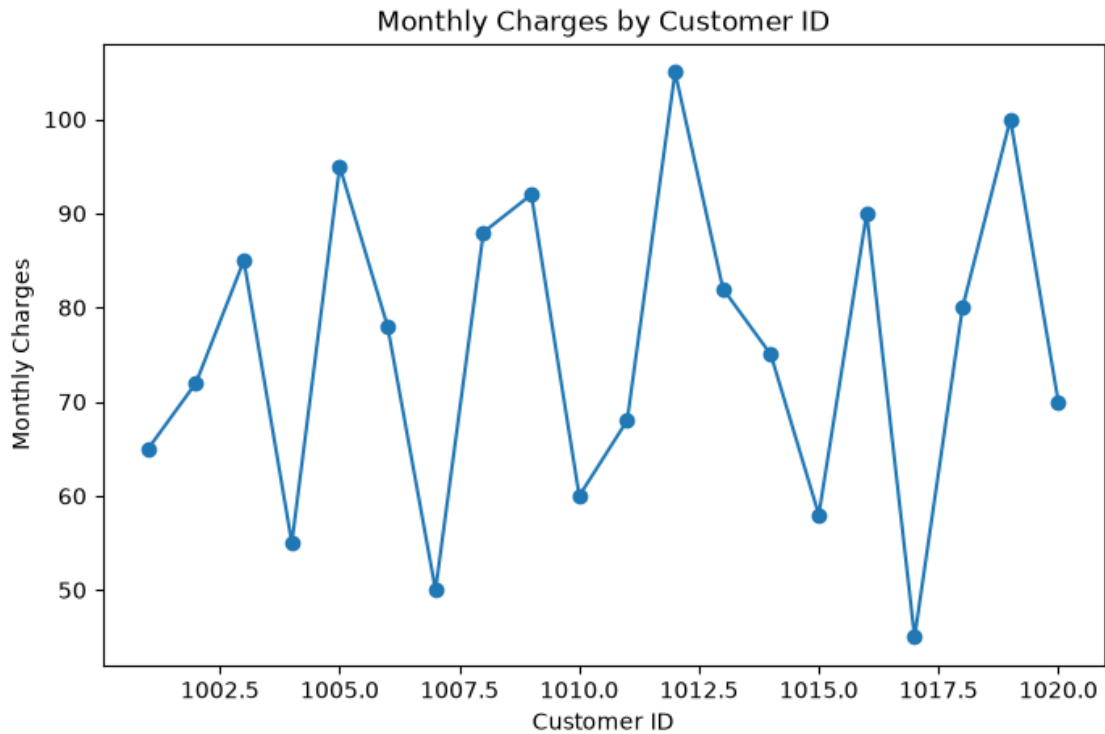
A Line Chart is used to understand trends or changes in a data. It Is useful when we want to know whether a value is increasing,decreasing, or a remaining stable. In this task, a line chart is used to visualize Monthly Charges for each Customer ID. This helps us observe the variation in Monthly Charges across customers.

```
[3]: plt.figure(figsize=(8, 5))

plt.plot(
    df["CustomerID"],
    df["MonthlyCharges"],
    marker="o"
)

plt.title("Monthly Charges by Customer ID")
plt.xlabel("Customer ID")
plt.ylabel("Monthly Charges")

plt.show()
```



0.4 2. Bar Chart

A Bar Chart is used to compare values across different categories. It helps us identify which category has a higher or lower number of the values. In this task, a bar chart is used to compare the number of the customers in different Contract types.

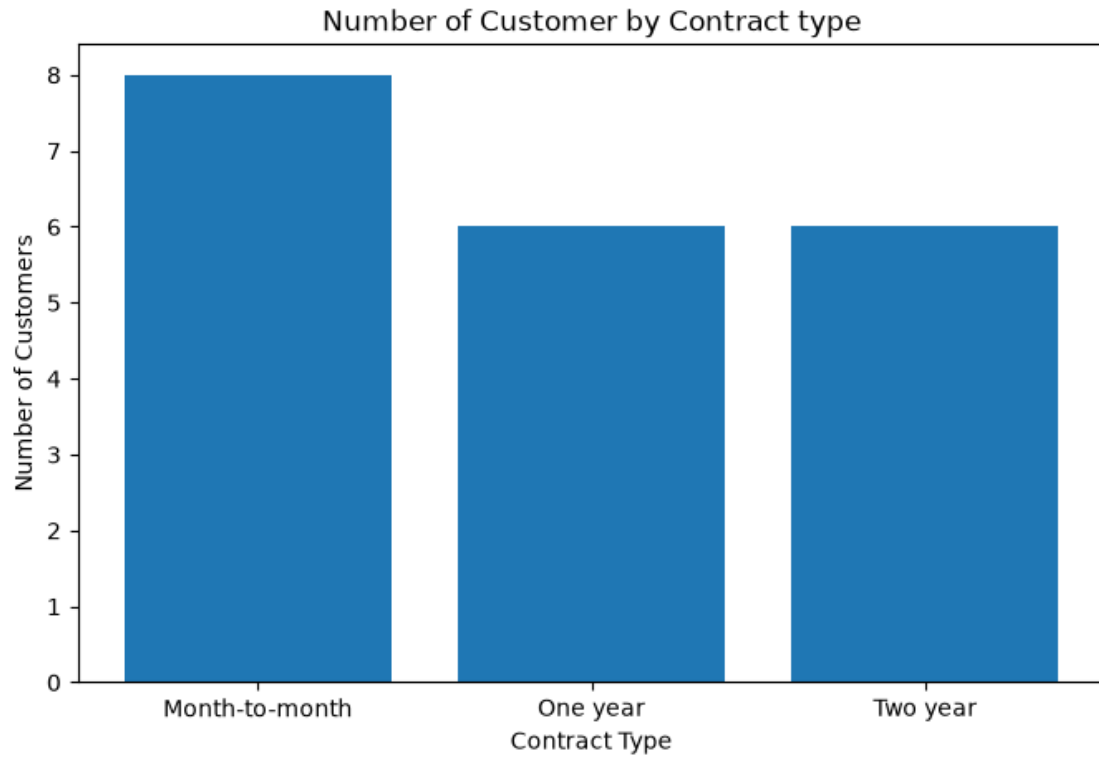
```
[4]: contract_counts = df["Contract"].value_counts()

plt.figure(figsize=(8,5))

plt.bar(
    contract_counts.index,
    contract_counts.values
)

plt.title("Number of Customer by Contract type")
plt.xlabel("Contract Type")
plt.ylabel("Number of Customers")

plt.show()
```



0.5 3. Histogram

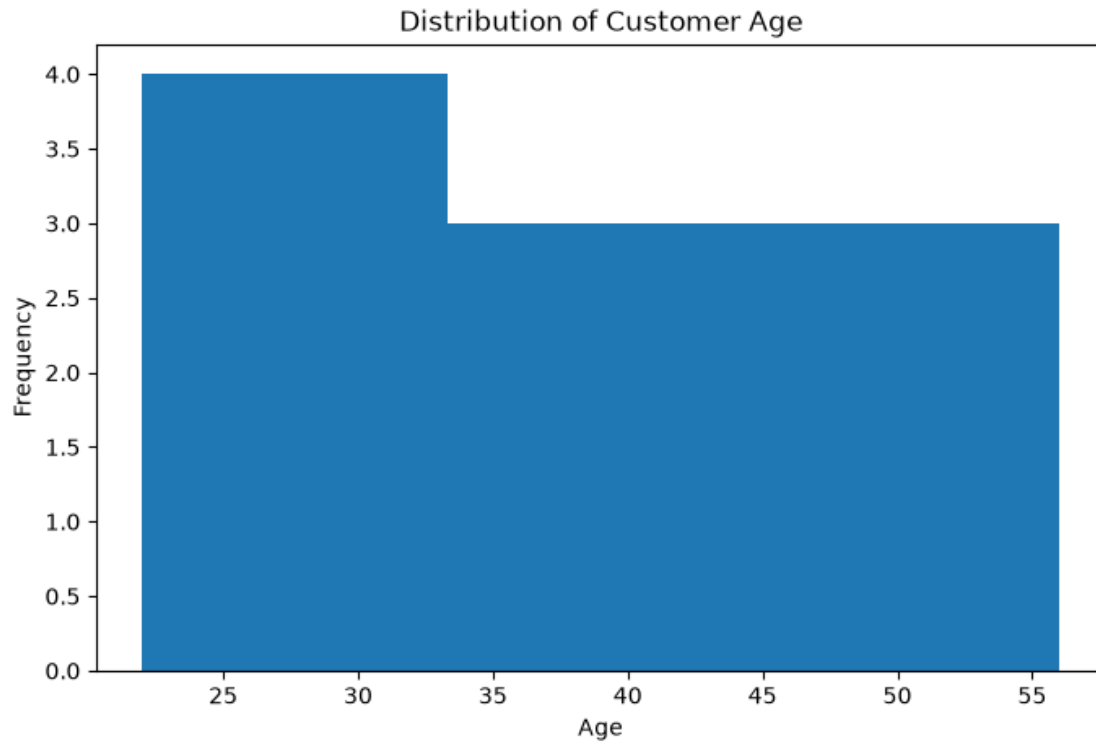
A Histogram is used to understand the distribution of numerical data. It shows how frequently values occur within different ranges. In this task, a histogram is used to visualize the distribution of the customer age.

```
[5]: plt.figure(figsize=(8,5))

plt.hist(
    df["Age"],
    bins=6
)

plt.title("Distribution of Customer Age")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```



0.6 4. Scatter Plot

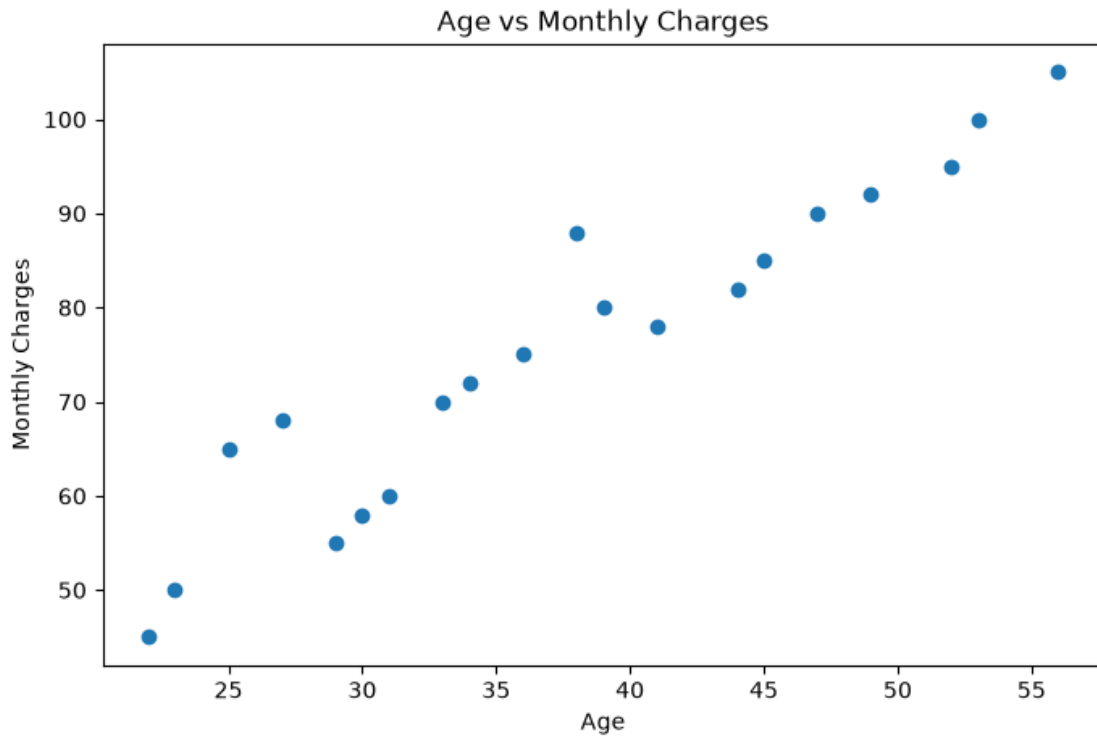
A Scatter Plot Is used to understand the relationship between two numerical variables. It helps us observe whether two variables have a positive,negative,or weak relationship. In this task, a scatter plot is used to visualize the relationships between Age and Monthly Charges.

```
[7]: plt.figure(figsize=(8,5))

plt.scatter(
    df["Age"],
    df["MonthlyCharges"]
)

plt.title("Age vs Monthly Charges")
plt.xlabel("Age")
plt.ylabel("Monthly Charges")

plt.show()
```



0.7 5. Count Plot

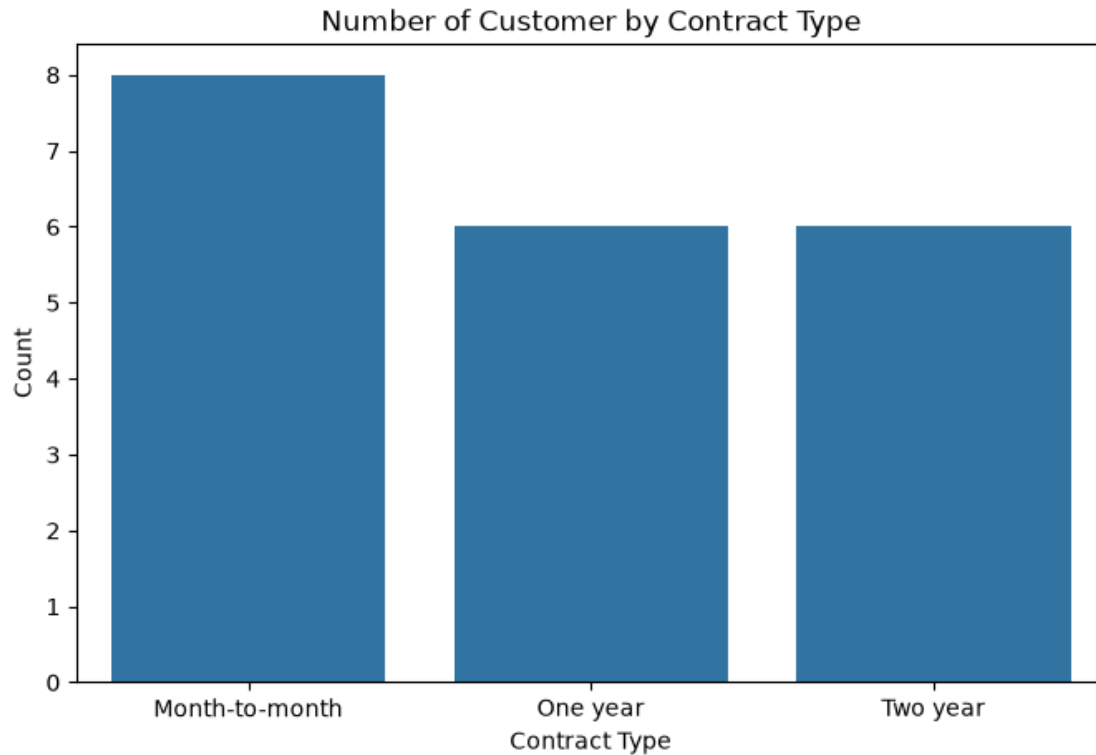
A Count Plot Is used to show the frequency or count of categorical values. It helps us understand how many records belong to each category. In this task, a count plot is used to show the number of customers in each Contract type.

```
[8]: plt.figure(figsize=(8,5))

sns.countplot(x=df["Contract"])

plt.title("Number of Customer by Contract Type")
plt.xlabel("Contract Type")
plt.ylabel("Count")

plt.show()
```



0.8 6. Heatmap

A Heatmap is used to visualize the correlation between numerical variables. It represents the strength of relationships between variables using a different shades. The correlation value ranges from -1 to +1. In this task, a heatmap is created for the numerical columns of the Customer Churn Dataset.

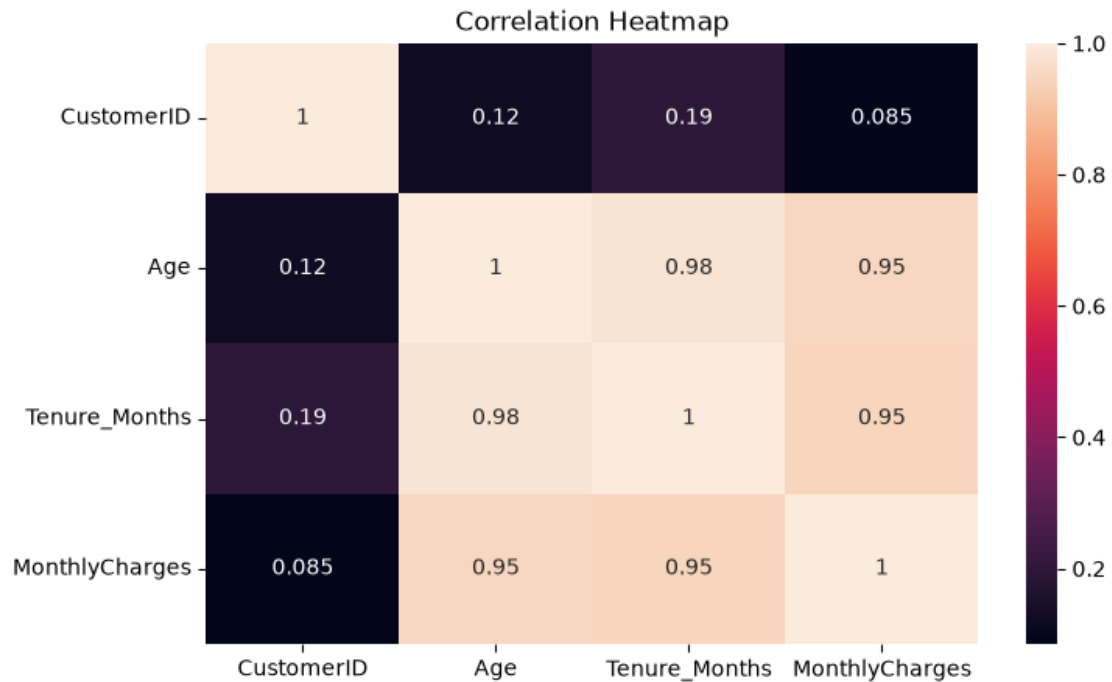
```
[9]: correlation = df.corr(numeric_only=True)

plt.figure(figsize=(8,5))

sns.heatmap(
    correlation,
    annot=True
)

plt.title("Correlation Heatmap")

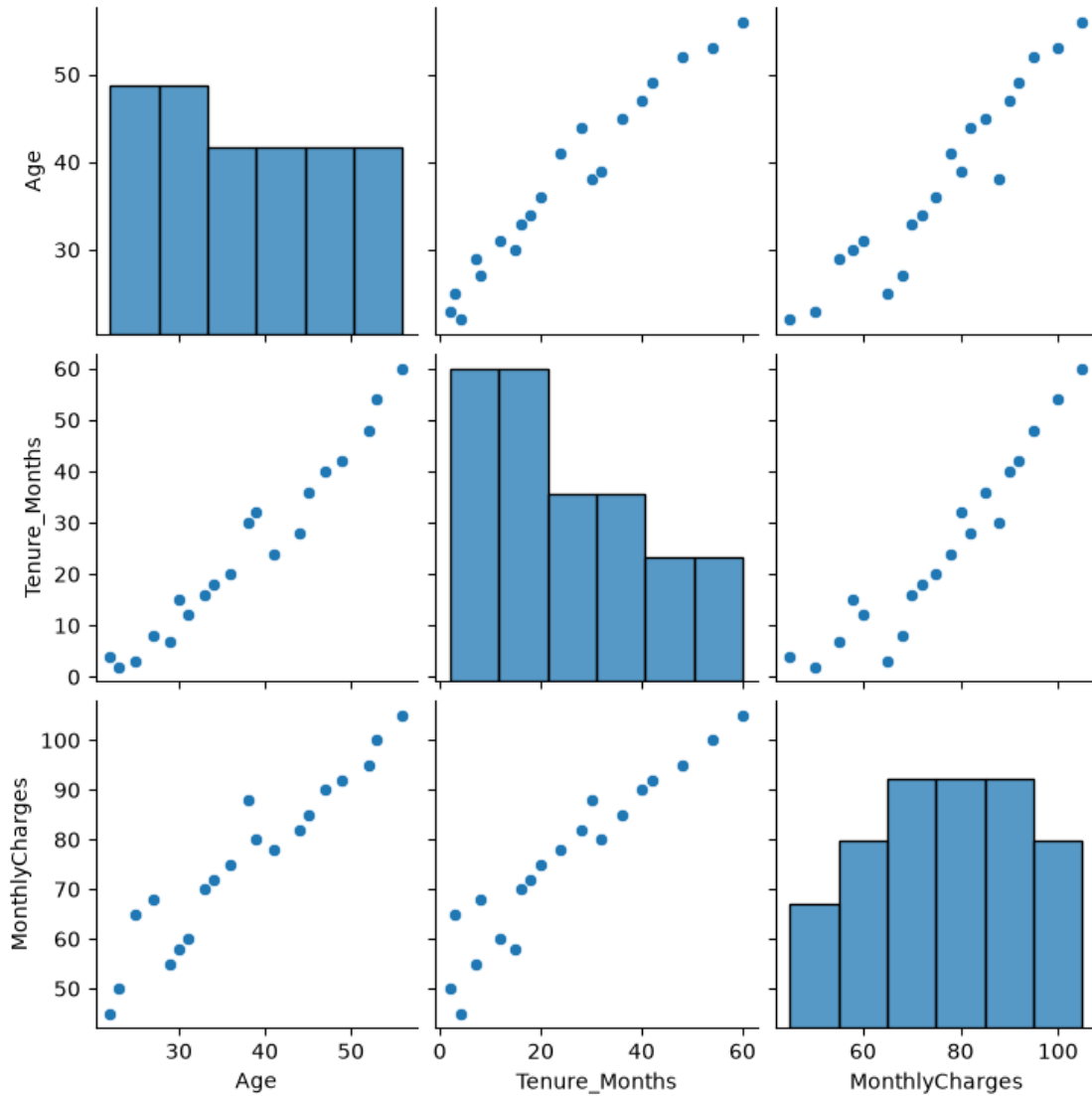
plt.show()
```



0.9 7. Pair Plot

A Pair Plot is used to visualize relationships between multiple numerical variables. It creates scatter plots for a pairs of a variables and distributions for a individual variables. In this task, a pair plot is created using Age,Tenure_Months, and MonthlyCharges.

```
[10]: sns.pairplot(  
    df[  
        [  
            "Age",  
            "Tenure_Months",  
            "MonthlyCharges"  
        ]  
    ]  
)  
  
plt.show()
```



0.10 Conclusion

In this task, seven different chart types were selected according to the questions being asked from the data.

The Line Chart was used to understand trends, the Bar Chart was used for comparison, the Histogram was used to understand data distribution, and the Scatter Plot was used to identify relationships between two numerical variables.

The Count Plot was used to show category frequency, the Heatmap was used to visualize correlation, and the Pair Plot was used to understand relationships between multiple numerical variables.

Choosing the right chart makes data analysis easier and helps communicate insights effectively.